UNIT-V

LINEAR ORDINARY DIFFERENTIAL EQUATIONS OF HIGHER ORDER

Topic Learning Objectives:

With Completion of this unit, students will be able to:

- Understand the concept of higher-order linear ordinary differential equations and their general form.
- Classify linear differential equations as homogeneous or non-homogeneous.
- Solve higher-order linear ODEs with constant coefficients using standard methods (complementary function and particular integral).
- Apply the method of variation of parameters to find particular solutions.
- Analyze the role of initial and boundary conditions in determining the unique solution of a higher-order ODE.
- Model and interpret real-world problems (mechanical vibrations, electrical circuits, control systems, etc.) using higher-order ODEs.
- Develop problem-solving skills by applying step-by-step procedures to obtain complete solutions.

Introduction:

Differential equations are used to express physical phenomena mathematically, and a single equation can often model a wide variety of processes. Differential equations are fundamental in mechanical engineering, enabling the modeling and analysis of dynamic systems and processes. Differential equations, particularly second-order ordinary differential equations (ODEs), are used to model the motion of vibrating systems like springs, the free, underdamped and critically damped motion, beams, and other mechanical components. They are used to model heat transfer in mechanical systems, such as heat exchangers, engines, and electronic devices.

Similar to this, LRC series circuits can be described by a linear second- order differential equation.

Differential Equation:

An equation containing the derivatives of one or more dependent variables, with respect to one or more independent variables, is said to be differential equation.

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The 'order' of a differential equation is the order of the highest derivative in the equation and the power of highest order derivative after clearing fractional powers from derivatives in the equation gives the 'degree' of differential equation.

Classification of differential equation:

Differential equations are classified as:

i) Ordinary differential equation (ODE)

ii) Partial differential equations (PDE)

An ordinary differential equation is that in which all the differential coefficients have reference to a single independent variable.

Examples:

i) $\ddot{x} = -n^2 x$

ii) $\frac{[1+(y')^2]^{3/2}}{y''} = a$

A partial differential equation is that in which there are two or more independent variables and partial differential coefficients with respect to any of them.

Examples:

i) $u_{xx} + u_{yy} = 0$

ii) $c^2 u_{xx} = u_t$

Linear Differential Equation:

A differential equation is said to be linear if dependent variable, all its derivatives are of first degree and they are not multiplied together.

A non-linear differential equation is one that is not linear.

Linear ODE of higher order:

A linear ODE of nth order is of the form:

$(a_0 D^n + a_1 D^{n-1} + \cdots + a_{n-1} D + a_n) y = g(x) \quad \ldots (1),$

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where $a_0, a_1, \cdots a_n$ are either constants or function of 'x' alone, $D = \frac{d}{dx}$ differential operator and $g(x)$ is a known function of 'x'.

Let $F(D) = (a_0 D^n + a_1 D^{n-1} + \cdots + a_{n-1} D + a_n)$ be a differential operator, then above equation can be written as:

$$F(D)y = g(x) \qquad \ldots (2)$$

If $g(x) = 0$, equation is said to be 'homogeneous' otherwise it is known as 'non-homogenous'.

If $y_1, y_2, \cdots, y_n$ are linearly independent solutions of the nth order homogeneous differential equation then

$$y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n,$$

where $c_1, c_2, \cdots, c_n$ are arbitrary constants is also a solution of homogeneous equation.

If y_p be a particular function which satisfies non-homogeneous linear differential equation then the general solution of the equation (2) is given by:

$$y = y_c + y_p,$$

where $y_c = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$ is called as complementary function and y_p is called particular integral.

Homogeneous Linear ODE with constant coefficients:

Consider a nth order homogeneous ODE,

$$(a_0 D^n + a_1 D^{n-1} + \cdots + a_{n-1} D + a_n)y = 0$$

Let $y = e^{mx}$ be a trial solution, obtaining the derivatives and using above equation we get:

$$e^{mx}(a_0 m^n + a_1 m^{n-1} + \cdots + a_{n-1} m + a_n) = 0$$

Since $e^{mx} \neq 0$, above equation vanishes if 'm' is root of algebraic equation. This algebraic equation is known as 'auxiliary' or 'characteristic' equation. The general solution of homogeneous equation depends on nature of roots of this equation.

Consider the second order ODE of the form:

$$(a_0 D^2 + a_1 D + a_n)y = 0$$

The auxiliary equation is given by $a_0 m^2 + a_1 m + a_n = 0$.

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Case (i) Let the roots of auxiliary equation be real and distinct say $m = m_1, m_2$. The general solution of above equation is given by $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

Case (ii) Let the roots of auxiliary equation be real and repeated say $m = m_1, m_1$. The general solution of above equation is given by $y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$

Case (iii) Let the roots of auxiliary equation be complex say $m = \alpha + i\beta, m = \alpha - i\beta$. The general solution of above equation is given by $y = e^{\alpha x}(c_1 cos\beta x + c_2 sin\beta x)$

Case (iv) If the ODE is fourth order and the roots of auxiliary equation are complex repeated say $m = \alpha \pm i\beta, m = \alpha \pm i\beta$. The general solution of above equation is given by

$$y = e^{\alpha x}[(c_1 + c_2 x)cos\beta x + (c_3 + c_4 x)sin\beta x]$$

The above procedure can be extended to higher order linear homogenous ODE in a similar manner.

Examples:

1. Solve: $2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 3y = 0$

Solution: The auxiliary equation is given by $2m^2 - 5m - 3 = 0$.

The roots are $m = -\frac{1}{2}, m = 3$.

The roots are real and distinct the general solution of given ODE is

$$y = c_1 e^{-(\frac{1}{2})x} + c_2 e^{3x}.$$

2. Solve: $\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 0$

Solution: The auxiliary equation is given by $(m-5)^2 = 0$

The roots are $m = 5, 5$ as the roots are real and repeated the general solution of given ODE is

$$y = c_1 e^{5x} + c_2 x e^{5x}$$

3. Solve: $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 7y = 0$

Solution: The auxiliary equation is given by $m^2 + 4m + 7 = 0$

The roots are $m = -2 \pm \sqrt{3}i$ which are complex, hence the general solution is given by

$$y = e^{-2x}[c_1 \cos(\sqrt{3}x) + c_2 \sin(\sqrt{3}x)]$$

4. Solve: $(D^3 + 1)y = 0$

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Solution: The auxiliary equation is given by $m^3 + 1 = 0$

The roots are $m = -1, \frac{1}{2} \pm i\frac{\sqrt{3}}{2}$, hence the general solution is given by

$$y=c_1e^{-x}+e^{\left(\frac{1}{2}\right)x}\left[c_1\cos\left(\frac{\sqrt{3}}{2}x\right)+c_2\sin\left(\frac{\sqrt{3}}{2}x\right)\right]$$

5. Solve: $(D^4 + 8D^2 + 16)y = 0$

Solution: The auxiliary equation is given by $m^4 + 8m^2 + 16 = 0$,

$$(m^2 + 4)^2 = 0$$

The roots are complex repeated $m = \pm 2i, \pm 2i$, hence the general solution is given by

$$y = [(c_1 + c_2x)\cos(2x) + (c_3 + c_4x)\sin(2x)]$$

6. Solve the initial value problem $\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 6x = 0$, given that $x(0) = 0, \frac{dx}{dt}(0) = 15$.

Solution: The auxiliary equation is given by $m^2 + 5m + 6 = 0$.

The roots are $m = -3, -2$, hence the general solution is given by $x = c_1e^{-3t} + c_2e^{-2t}$

Using the given initial conditions, we find that $c_1 + c_2 = 0$ and $2c_1 + 3c_2 = -15$.

Solving these we get $c_1 = 15$ and $c_2 = -15$.

The particular solution of the given differential equation is

$x = 15(e^{-3t} - e^{-2t})$

7. Solve the initial value problem $2x''(t) + 5x'(t) - 3x(t) = 0$, given that $x(0) = 1, x'(0) = 4$.

Solution: The auxiliary equation is given by $2m^2 + 5m - 3 = 0$. The roots are $m = -3, \frac{1}{2}$,

hence the general solution is given by $x = c_1e^{\frac{1}{2}t} + c_2e^{-3t}$.

Using the given initial conditions, we find that $c_1 + c_2 = 1$ and $c_1 - 6c_2 = 8$.

Solving these we get $c_1 = 2$ and $c_2 = -1$.

The particular solution of the given differential equation is

$$x = 2e^{\frac{1}{2}t} - e^{-3t}.$$

8. Obtain the solution of the initial value problem $y''(x) - 4y'(x) + 4y(x) = 0$, given that $y(0) = 0, y'(0) = 3$.

Solution: The auxiliary equation is given by

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$m^2 - 4m + 4 = 0$. The roots are $m = 2, 2$, hence the general solution is given by

$$y = c_1 e^{2x} + c_2 x e^{2x}.$$

Using the given initial conditions, $c_1 = 0$ and $c_2 = 3$.

The particular solution of the given differential equation is

$$y = 3xe^{2x}.$$

Exercise:

1 Solve $(D^3 - 8)y = 0$, where $D \equiv \frac{d}{dx}$.

Answer: $y = c_1 e^{2x} + e^{-x}\left(c_2 \cos\sqrt{3}\,x + c_3 \sin\sqrt{3}\,x\right)$.

2 Solve $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 0; y(0) = 0, \frac{dy(0)}{dt} = 15$.

Answer: $y = 15(e^{-2t} - e^{-3t})$.

3 Solve $(D^4 - 4D^3 + 8D^2 - 8D + 4)y = 0$, where $D \equiv \frac{d}{dx}$.

Answer: $y = e^x[(c_1 + c_2 x)\cos x + (c_3 + c_4 x)\sin x]$.

4 Solve $(D^3 - D^2 - D + 1)y = 0$, where $D \equiv \frac{d}{dx}$.

Answer: $y = (c_1 + c_2 x)e^x + c_3 e^{-x}$.

5 Solve $\frac{d^3y}{dx^3} - 2\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 8y = 0$.

Answer: $y = c_1 e^{2x} + c_2 \cos 2x + c_3 \sin 2x$.

6 Find the solution of the boundary value problem

$y'' + y = 0$ given that $y(0) = 1, y\left(\frac{\pi}{2}\right) = 2$.

Answer: $y = \cos x + 2\sin x$.

7 Find the solution of the initial value problem:

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 0; y(0) = 0, \frac{dy(0)}{dt} = 15.$$

Answer: $y = 15(e^{-2t} - e^{-3t})$.

Non-homogeneous Linear ODE with constant coefficients:

The general solution of non-homogeneous linear ODE has two parts –

Complementary function and Particular integral

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The complementary function is solution of homogeneous ODE.

The rules for finding particular integral are given below:

The kinds of functions that make up the input function $g(x)$ are generally constants, exponentials, sine, cosine and polynomials.

Consider F(D)y = g(x) ,

If $y = y_p$ is a particular integral of this ODE, then

$F(D)y_p = g(x)$, then $y_p = \frac{1}{F(D)} g(x)$,

where $\frac{1}{F(D)}$ is called as 'inverse differential operator'.

Rules for finding particular integral:

i) Let $g(x) = ke^{ax}$

then $y_p = k\frac{1}{F(D)}e^{ax} = k\frac{1}{F(a)}e^{ax}$, provided $F(a) \neq 0$

If $F(a) = 0$ then $y_p = k\frac{x}{[F'(D)]_{D=a}}e^{ax}$, provided $F'(a) \neq 0$

If $F'(a) = 0$ then $y_p = k\frac{x^2}{[F''(D)]_{D=a}}e^{ax}$, provided

$F''(a) \neq 0$ and so on

ii) Let $g(x) = \sin(ax+b)$ or $\cos(ax+b)$

$$y_p = \frac{1}{F(D^2)}\sin(ax+b) \; or \; \frac{1}{F(D^2)}\cos(ax+b)$$

$$= \frac{1}{F(-a^2)}\sin(ax+b) \; or \; \frac{1}{F(-a^2)}\cos(ax+b), \text{ provided } F(-a^2) \neq 0.$$

If $F(-a^2) = 0$, $y_p = \frac{x}{F'(-a^2)}\sin(ax+b) \; or \; \frac{x}{F'(-a^2)}\cos(ax+b)$,

provided $F'(-a^2) \neq 0$

Particular case: $\frac{1}{(D^2+a^2)}\cos(ax+b) = \frac{x}{2a}\sin(ax+b)$

$$\frac{1}{(D^2+a^2)}\sin(ax+b) = -\frac{x}{2a}\cos(ax+b)$$

iii) Let $g(x) = x^m$ then $y_p = \frac{1}{F(D)}x^m = [F(D)]^{-1}x^m$

Expanding the right hand side as a binomial series, the particular integral can be obtained. The following series expansions are useful:

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$(1+D)^{-1} = 1 - D + D^2 - D^3 + \cdots$

$(1-D)^{-1} = 1 + D + D^2 + D^3 + \cdots$

$(1+D)^{-2} = 1 - 2D + 3D^2 - 4D^3 + \cdots$

$(1-D)^{-2} = 1 + 2D + 3D^2 + 4D^3 + \cdots$

iv) Let $g(x) = e^{ax}V(x)$ where $V(x)$ is function of 'x'

$$y_p = \frac{1}{F(D)} e^{ax}V(x) = e^{ax}\frac{1}{F(D+a)}V(x)$$

The right hand side can be evaluated by earlier methods.

Examples:

1. Solve: $(D-2)^2 y = 8(e^{2x} + sin2x + x^2)$

Solution: The auxiliary equation is $(m-2)^2 = 0$, the roots are $m = 2, 2$

$\therefore y_c = c_1 e^{2x} + c_2 x e^{2x}$

The particular integral is obtained as follows:

$$y_p = 8\left[\frac{1}{(D-2)^2}e^{2x} + \frac{1}{(D-2)^2}sin2x + \frac{1}{(D-2)^2}x^2\right]$$

Now $\frac{1}{(D-2)^2}e^{2x} = x^2\frac{1}{[2(1-0)]_{D=2}}e^{2x} = \frac{x^2e^{2x}}{2}$

$$\frac{1}{(D-2)^2}sin2x = \frac{1}{(D^2-D+4)}sin2x = \frac{1}{-4-4D+4}\sin 2x$$

$$=-\frac{1}{4D}\sin 2x = -\frac{1}{4}\int \sin(2x)\,dx = \frac{1}{8}\cos 2x$$

$$\frac{1}{(D-2)^2}x^2 = \frac{1}{4}\left(-\frac{D}{2}\right)^{-2}x^2 = \frac{1}{4}\left(1+\frac{2D}{2}+\frac{3D^2}{4}+\ldots\right)x^2$$

$$=\frac{1}{4}\left(x^2+2x+\frac{3}{2}\right)$$

$\therefore y_p = 4x^2e^{2x} + cos2x + 2x^2 + 4x + 3$

Hence, the general solution of given equation is

$y = y_c + y_p$

$$y = c_1e^{2x} + c_2xe^{2x} + 4x^2e^{2x} + cos2x + 2x^2 + 4x + 3$$

2. Solve: $y'' - 4y = \cosh(2x-1) + 3^x$

Solution: The auxiliary equation and its roots are:

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$m^2 - 4 = 0, m = +2, -2.$

$\therefore y_c = c_1 e^{-2x} + c_2 e^{2x}$

The particular integral is obtained as follows:

$$y_p = \frac{1}{(D^2-4)}\cosh(2x-1) + \frac{1}{(D^2-4)}3^x$$

Now $\frac{1}{(D^2-4)}\cosh(2x-1) = \frac{1}{2}\left[\frac{e^{2x-1}+e^{-2x+1}}{D^2-4}\right]$

$$= \frac{1}{8}(xe^{2x-1} - xe^{-2x+1})$$

$$\frac{1}{(D^2-4)}3^x = \frac{1}{(D^2-4)}e^{xlog(3)} = \frac{1}{[(log3)^2-4]}3^x$$

Hence, the solution of given equation is $y = y_c + y_p$

$\therefore y = c_1 e^{-2x} + c_2 e^{2x} + \frac{1}{8}(xe^{2x-1} - xe^{-2x+1}) + \frac{1}{[(\log 3)^2-4]}3^x$

3. Solve: $y'' - 2y' + y = xe^x sinx$

Solution: The auxiliary equation and its roots are:

$$m^2 - 2m + 1 = 0, m = 1, 1.$$

$$\therefore y_c = c_1 e^x + c_2 x e^x$$

The particular integral is obtained as follows:

$$y_p = \left[\frac{1}{(D-1)^2}\right] e^x xsinx = e^x \frac{1}{[D+1-1]^2} xsinx$$

$$= e^x \frac{1}{D^2} xsinx$$

Integrating right hand side twice by using the rule of parts we obtain:

$y_p = -e^x(xsinx + 2cosx)$

Hence, the solution of given equation is $y = y_c + y_p$

$\therefore y = c_1 e^x + c_2 x e^x - e^x(xsinx + 2cosx)$

4. Solve: $(D^3 + 3D^2 + 3D + 1)y = e^{-x}$

Solution: The auxiliary equation and its roots are:

$m^3 + 3m^2 + 3m + 1 = 0, m = -1, -1, -1.$

$\therefore y_c = c_1 e^{-x} + c_2 x e^{-x} + c_3 x^2 e^{-x}$

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Now $\frac{1}{(D+1)^3}e^{-x} = x^2\frac{1}{[6(D+1)]_{D=-1}}e^{-x} = \frac{x^3e^{-x}}{6}$

Hence, the solution of given equation is $y = y_c + y_p$

$\therefore y = c_1e^{-x}+c_2xe^{-x}+c_3x^2e^{-x}+\frac{x^3e^{-x}}{6}$

5. Find the particular integral of

$(D^3+1)y = \cos(2x-1)$.

Solution: The particular integral is given by:

$$y_p = \left[\frac{1}{D^3+1}\right]\cos(2x-1) = \frac{1}{D(-4)+1}\cos(2x-1)$$

$$= \frac{(1+4D)}{1-16D^2}\cos(2x-1)$$

$$y_p = \frac{(1+4D)}{1-16(-4)}\cos(2x-1) = \frac{1}{65}[\cos(2x-1) - 8\sin(2x-1)]$$

6. Find the particular integral of $(D^3+4D)y = 2\sin x\cos x$.

Solution: The particular integral is given by:

$$y_p = \left[\frac{1}{D^3+4D}\right]2\sin x\cos x = \frac{1}{(D^2+4)D}\sin(2x)$$

$$= x\frac{1}{3D^2+4}\sin(2x) = x\frac{1}{3(-4)+4}\sin(2x)$$

$$= -\frac{x}{8}\sin(2x)$$

7. Find the particular integral of $(D^2+9)y = x\cos x$

Solution: The particular integral is given by:

$$y_p = \frac{1}{D^2+9}Re(xe^{ix}) = Re\frac{1}{D^2+9}(xe^{ix})$$

$$= Re\, e^{ix}\frac{1}{(D+i)^2+9}(\mathrm{x}) = Re\, e^{ix}\frac{1}{D^2+2iD+8}(\mathrm{x})$$

$$= Re\, e^{ix}\frac{1}{8\left(1+\frac{iD}{4}\right)}(\mathrm{x}) = Re\frac{e^{ix}}{8}\left(1+\frac{iD}{4}\right)^{-1}(\mathrm{x})$$

$$= Re\frac{e^{ix}}{8}\left(1-\frac{iD}{4}\right)(\mathrm{x}) = Re\frac{\cos x+i\sin x}{8}\left(x-\frac{i}{4}\right)$$

$$= \frac{1}{32}(4x\cos x+\sin x)$$

8. Find the particular integral of $(D^2-4D+4)y = x^2e^{3x}+\sin^2 x$

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Solution: The particular integral is given by:

$$y_p = \frac{1}{(D-2)^2}[x^2e^{3x} + \sin^2 x]$$

$$= \frac{1}{(D-2)^2}[x^2e^{3x}] + \frac{1}{(D-2)^2}[\sin^2 x]$$

Now, $\frac{1}{(D-2)^2}[x^2e^{3x}] = e^{3x}\frac{1}{(D+3-2)^2}[x^2]$

$$= e^{3x}\frac{1}{(D+1)^2}[x^2]$$

$$= e^{3x}(1+D)^{-2}[x^2]$$

$$= e^{3x}(1-2D+3D^2)[x^2]$$

$$= e^{3x}(x^2-4x+6)$$

and $\frac{1}{(D-2)^2}[\sin^2 x] = \frac{1}{(D-2)^2}\left[\frac{1-\cos 2x}{2}\right]$

$$= \frac{1}{(D-2)^2}\left[\frac{1}{2}e^{0x}\right] - \frac{1}{(D-2)^2}\left[\frac{1}{2}\cos 2x\right]$$

$$= \frac{1}{(0-2)^2}\left[\frac{1}{2}e^{0x}\right] - \frac{1}{D^2-4D+4}\left[\frac{1}{2}\cos 2x\right]$$

$$= \frac{1}{8} - \frac{1}{2}.\frac{1}{2^2-4D+4}[\cos 2x]$$

$$= \frac{1}{8} + \frac{1}{8D}[\cos 2x] = \frac{1}{8} + \frac{1}{16}\sin 2x$$

$$\therefore y_p = e^{3x}(x^2-4x+6) + \frac{1}{8} + \frac{1}{16}\sin 2x$$

9. Find the general solution of $y'' - 4y = xe^x$.

Solution: The auxiliary equation and its roots are:

$m^2 - 4 = 0, m = +2, -2.$

$\therefore y_c = c_1e^{-2x} + c_2e^{2x}.$

Now, $y_p = \frac{1}{(D^2-4)}[xe^x] = e^x\frac{1}{(D+1)^2-4}[x]$

$$= e^x\frac{1}{(D^2+2D-3)}[x]$$

$$= \frac{-e^x}{3}\left(1 - \frac{(D^2+2D)}{3}\right)^{-1}[x]$$

$$= \frac{-e^x}{3}\left(1 + \frac{(D^2+2D)}{3}\right)[x]$$

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$$= \frac{-e^x}{3}\left(x + \frac{2}{3}\right)$$

The complete solution is $y = y_c + y_p$

$y = c_1 e^{-2x} + c_2 e^{2x} - \frac{e^x}{3}\left(x + \frac{2}{3}\right)$.

10. Solve $(D^3 + 1)y = e^{-x} + sinx$.

Solution: The auxiliary equation and its roots are:

$m^3 + 1 = 0 \Rightarrow m = -1, m = \frac{1 \pm i\sqrt{3}}{2}$

$\therefore y_c = c_1 e^{-x} + e^{\frac{1}{2}x}(c_2 \cos\frac{\sqrt{3}}{2}x + c_3 \sin\frac{\sqrt{3}}{2}x)$

Now $y_p = \frac{1}{(D^3+1)}e^{-x} + \frac{1}{(D^3+1)}\sin x = \frac{xe^{-x}}{3D^2} + \frac{\sin x}{1-D}$

$y_p = \frac{xe^{-x}}{3} + \frac{(1+D)}{(1-D^2)}.\sin x = \frac{xe^{-x}}{3} + \left(\frac{\sin x + \cos x}{2}\right)$

The complete solution is $y = y_c + y_p$

$y = c_1 e^{-x} + e^{\frac{1}{2}x}(c_2 \cos\frac{\sqrt{3}}{2}x + c_3 \sin\frac{\sqrt{3}}{2}x) + \frac{xe^{-x}}{3} + \left(\frac{\sin x + \cos x}{2}\right)$.

Exercise:

1. Solve $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 2e^{2t}$.

   Answer: $y = (c_1 + c_2 t)e^{-t} + \frac{2}{9}e^{2t}$

2. Solve $y''' - 12y' + 16y = (e^x + e^{-2x})^2$.

   Answer: $y = (c_1 + c_2 x)e^{2x} + c_3 e^{-4x} + \frac{x^2}{12}e^{2x} + \frac{2}{27}e^{-x} + \frac{x}{36}e^{-4x}$

3. Solve $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} - \frac{dy}{dx} - y = cos\,2\,x$.

   Answer: $y = ae^x + (b + cx)e^{-x} - \frac{1}{25}(2\,sin\,2\,x + cos\,2\,x)$

4. Solve $(D^3 - D^2 - 6D)x = 1 + t^2$.

   Answer: $x = a + be^{3t} + ce^{-2t} - \frac{1}{108}(25t - 3t^2 + 6t^3)$

5. Solve $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{2x}\,sin\,x$.

   Answer: $y = ae^{-x} + be^{-2x} - \frac{1}{170}e^{2x}(7\,cos\,x - 11\,sin\,x)$

6. Solve $(D^4 + 4)y = sin^2\,x$.

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Answer: $y = a\cos 2x + b\sin 2x + \frac{1}{8}(1 - x\sin 2x)$

7. Solve $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = e^{3x}(x^2 + 7x + 5)$.

Answer: $y = \left(a + bx + \frac{1}{12}(x^4 + 14x^3 + 30x^2)\right)e^{3x}$

8. Solve $\frac{d^2y}{dx^2} - 4y = x\sinh x$.

Answer: $y = ae^{2x} + be^{-2x} - \frac{x}{3}\sinh x - \frac{2}{9}\cosh x$

9. Solve $(D^2 - 4D + 4)y = 8x^2e^{2x}\sin 2x$.

Answer: $y = (a + bx(3 - 2x^2)\sin 2x - 4x\cos 2x)e^{2x}$

10. Solve $y'' - y = x\sin x + (1 + x^2)e^x$.

Answer: $y = c_1e^{-x} + c_2e^x - \frac{x}{2}\sin x - \frac{1}{2}\cos x + \frac{e^x}{12}\left(\frac{3x}{2} - \frac{x^2}{2} + \frac{x^3}{3}\right)$.

Method of Variation of Parameters:

The method of variation of parameter is more general method, in which the function g(x) is not restricted to any particular form. The working rule for finding solution of the given ODE is given below:

Consider the second order ODE of the form

$y'' + P(x)y' + Q(x)y = g(x)$

Let $y = c_1y_1 + c_2y_2$ be solution of the equation with $g(x) = 0$, the complete solution of given ODE is

$y = A(x)y_1 + B(x)y_2$, where

$A(x) = -\int \frac{y_2 g(x)}{W}dx + k_1$ and $B(x) = \int \frac{y_1 g(x)}{W}dx + k_2$

With the 'Wronskian' given by $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \neq 0$

Examples:

1. Solve: $y'' + a^2y = \sec(ax)$

Solution: The auxiliary equation and its roots are

$m^2 + a^2 = 0, m = \pm ai. \therefore y_c = c_1\cos(ax) + c_2\sin(ax)$

Let $y_1 = \cos(ax)$, $y_2 = \sin(ax)$

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$$W=\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = a\left[\cos^2(ax)+\sin^2(ax)\right] = a$$

$$A(x)=-\int \frac{y_2 g(x)}{W}dx+k_1$$

$$=-\int \frac{\tan(ax)}{a}dx+k_1=\frac{1}{a^2}\log[\cos(ax)]+k_1$$

$$B(x)=\int \frac{y_1 g(x)}{W}dx+k_2=\frac{1}{a}\int dx+k_2=\frac{x}{a}+k_2$$

The complete solution of given ODE is

$$y = A(x)y_1 + B(x)y_2,$$

$$y = \frac{1}{a^2}\log[\cos(ax)]\cos(ax)+\frac{x}{a}\sin(ax).$$

2. Solve: $y'' - y = e^{-2x}\cos(e^{-x})$

Solution: The auxiliary equation and its roots are

$m^2 - 1 = 0, m = \pm 1. \therefore y_c = c_1 e^{-x} + c_2 e^{x}$

Let $y_1 = e^{x}$, $y_2 = e^{-x}$, $W=\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}=-2$

$$A(x) =-\int \frac{y_2 g(x)}{W}dx+k_1 =-\frac{1}{2}\int z^2\cos(z)\,dz+k_1,$$

where $z = e^{-x}$, and $dz = -e^{-x}dx$

Integrating by rule of parts obtain:

$$A(x) = -\frac{1}{2}e^{-2x}\sin(e^{-x}) - e^{-x}\cos(e^{-x}) + \sin(e^{-x}) + k_1 \quad B(x) = \int \frac{y_1 g(x)}{W}dx+k_2$$

$$=\frac{1}{2}\int \cos(z)dz+k_2=\frac{1}{2}\sin(e^{-x})+k_2, \text{ where } z= e^{-x}$$

The complete solution of given ODE is

$$y = A(x)y_1 + B(x)y_2$$

$$\therefore y= k_1e^{-x}+k_2e^{x}-\cos(e^{-x})+e^{x}\sin(e^{-x}).$$

3. Solve $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = \frac{e^x}{1+e^x}$ by the method of variation of parameters.

Solution: The auxiliary equation and its roots are

$m^2 - 3m + 2 = 0, m = 1, 2. \therefore y_c = c_1 e^{x} + c_2 e^{2x}$

Let $y_1 = e^{x}$, $y_2 = e^{2x}$, $W=\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}=e^{3x}$

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$A(x) = -\int \frac{y_2 g(x)}{W} dx + k_1 = -\int \frac{e^x . e^{2x}}{(1+e^x)e^{3x}} dx + k_1$,

$A(x) = -\int \frac{1}{(1+e^x)} dx + k_1 = -\int \frac{e^{-x}}{(1+e^{-x})} dx + k_1$

$= log(1 + e^{-x}) + k_1$

$B(x) = \int \frac{y_1 g(x)}{W} dx + k_2 = \int \frac{e^x . e^x}{(1+e^x)e^{3x}} dx + k_2$

$= \int \frac{e^{-x}}{(1+e^x)} dx + k_2 = \int \frac{e^{-2x}}{(1+e^{-x})} dx + k_2$

$= log(1 + e^{-x}) - (1 + e^{-x}) + k_2$

The complete solution of given ODE is

$y = A(x)y_1 + B(x)y_2$

$\therefore y = [log(1 + e^{-x}) + k_1]e^x + [log(1 + e^{-x}) - (1 + e^{-x}) + k_2]e^{2x}$.

4.Solve: $y'' + 4y = 4sec^2(2x)$ by the method of variation of parameters.

Solution: The auxiliary equation and its roots are

$m^2 + 4 = 0, m = \pm 2i. \therefore y_c = c_1 cos(2x) + c_2 sin(2x)$

Let $y_1 = \cos(2x)$, $y_2 = \sin(2x)$

$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos(2x) & \sin(2x) \\ -2\sin(2x) & 2\cos(2x) \end{vmatrix} = 2$

$A(x) = -\int \frac{y_2 g(x)}{W} dx + k_1 = -\int \frac{\sin(2x) 4sec^2(2x)}{2} dx + k_1$

$= -2\int \sec(2x)\tan(2x) dx + k_1 = -\sec(2x) + k_1$

$B(x) = \int \frac{y_1 g(x)}{W} dx + k_2 = \int \frac{\cos(2x) 4sec^2(2x)}{2} dx + k_2$

$= 2\int \sec(2x) dx + k_1 = \log[\sec(2x) + \tan(2x)] + k_2$

The complete solution of given ODE is

$y = A(x)y_1 + B(x)y_2$,

$y = [-\sec(2x) + k_1]\cos(2x) + [\log[\sec(2x) + \tan(2x)] + k_2]\sin(2x)$.

Exercise:

1. Using the method of variation of parameters, solve $(D^2 + 4)x = tan\, 2\, t$.

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Answer: $x = c_1 \cos 2t + c_2 \sin 2t - \frac{1}{4}\cos 2t \log(\sec 2t + \tan 2t)$

2. Solve $\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x}$ by the method of variation of parameters.

   Answer: $y = c_1 e^x + c_2 e^{-x} - 1 + e^x \log(e^{-x} + 1) - e^{-x} \log(e^x + 1)$.

3. Solve $y'' - 2y' + y = e^x \log x$ by the method of variation of parameters.

   Answer: $y = (c_1 + c_2 x)e^x + \frac{1}{4}x^2 e^x (2 \log x - 3)$.

4. Using the method of variation of parameters, solve $(D^2 - 1)y = 2(1 - e^{-2x})^{-1/2}$.

   Answer: $y = [-\sin^{-1}(e^{-x}) + c_1]e^x + \left[-(e^{2x} - 1)^{\frac{1}{2}} + c_2\right] e^{-x}$.

5. Solve $y'' - 2y' = e^x \sin x$ by the method of variation of parameters.

   Answer: $y = c_1 + c_2 e^{2x} - \frac{e^x}{2} \sin x$.

Cauchy-Euler equation:

The linear ODE of the form

$$(a_0 x^n D^n + a_1 x^{n-1} D^{n-1} + a_2 x^{n-2} D^{n-2} + \cdots + a_{n-1} x D + a_n)y = g(x)$$

where $a_0, a_1, \cdots a_n$ are constants, is known as 'Cauchy-Euler' or equidimensional equation.

This equation can be reduced to ODE with constant coefficients by changing the independent variable as follows –

Take $x = e^z$, then $xDy = D_1 y$,

$$x^2 D^2 y = D_1(D_1 - 1)y,$$

$$x^3 D^3 y = D_1(D_1 - 1)(D_1 - 2)y$$

where $D_1 = \frac{d}{dz}$ and so on.

The resulting ODE can be solved using the earlier methods.

Examples:

1. Solve: $4x^2 \frac{d^2y}{dx^2} + 8x \frac{dy}{dx} + y = 0$

Solution: The given ODE is second order Euler equation.

Put $x = e^z$, then $xDy = D_1 y$, $x^2 D^2 y = D_1(D_1 - 1)y$

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The given equation reduces to $({4D_1}^2 + 4D_1 + 1)y = 0$

The auxiliary equation and its roots are

$4m^2 + 4m + 1 = 0, m = -\frac{1}{2}, -\frac{1}{2}.$

The general solution of given ODE is

$y = c_1 e^{-(\frac{1}{2})z} + c_2 z e^{-(\frac{1}{2})z}$, where $z = \log(x)$.

2. Solve: $x^2 \frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = \log(x)$.

Solution: The given ODE is second order Euler equation.

Put $x = e^z$, then $xDy = D_1 y,\ x^2D^2y = D_1(D_1 - 1)y$

The given equation reduces to $({D_1}^2 - 2D_1 + 1)y = z$

The auxiliary equation and its roots are

$m^2 - 2m + 1 = 0,\ m = 1, 1.$

$\therefore y_c = c_1 e^z + c_2 z e^z$

Now $y_p = \frac{1}{(D_1-1)^2} z = (1 - D_1)^{-2} z = (1 + 2D_1 + 3{D_1}^2 + \ldots) z = z + 2$

Hence, the solution of given equation is $y = y_c + y_p$

$y = c_1 e^z + c_2 z e^z + z + 2$, where $z = \log(x)$.

3. Solve $x^2 \frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 5y = x^2 \sin(\log_e(x))$.

Solution: The given ODE is second order Euler equation.

Put $x = e^z$, then $xDy = D_1 y,\ x^2D^2y = D_1(D_1 - 1)y$

The given equation reduces to

$({D_1}^2 - 4D_1 + 5)y = e^{2z} sinz$

The auxiliary equation and its roots are

$m^2 - 4m + 5 = 0, m = 2 \pm i.$

$\therefore y_c = e^{2z}[c_1 cosz + c_2 sinz].$

Now $y_p = \frac{1}{{D_1}^2 - 4D_1 + 5} e^{2z} sinz$

$y_p = e^{2z} \frac{1}{(D_1+2)^2 - 4(D_1+2) + 5} sinz$

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$= e^{2z}\frac{1}{{D_1}^2+1}sinz = e^{2z}z\frac{1}{2D_1}sinz = \frac{-z}{2}e^{2z}cosz$

Hence, the solution of given equation is $y = y_c + y_p$

$y = e^{2z}[c_1cosz + c_2sinz] - \frac{z}{2}e^{2z}cosz$, where $z = \log(x)$.

4. Solve $x^2\frac{d^2y}{dx^2} + 4x\frac{dy}{dx} + 2y = e^x$.

Solution: The given ODE is second order Euler equation.

Put $x = e^z$, then $xDy = D_1y,\ x^2D^2y = D_1(D_1 - 1)y$

The given equation reduces to

$({D_1}^2 + 3D_1 + 2)y = e^{e^z}$

The auxiliary equation and its roots are

$m^2 + 3m + 2 = 0, m = -1, -2.$

$\therefore y_c = c_1e^{-z} + c_2e^{-2z}.$

Now $y_p = \frac{1}{({D_1}^2+3D_1+2)}e^{e^z} = \left(\frac{1}{D_1+1} - \frac{1}{D_1+2}\right)e^{e^z}$

$= \frac{1}{D_1+1}\left(e^{e^z}\right) - \frac{1}{D_1+2}\left(e^{e^z}\right)$

$= e^{-z}\int e^z.e^{e^z}\,dz - e^{-2z}\int e^{2z}.e^{e^z}\,dz$

$= e^{-z}.e^{e^z} - e^{-2z}(e^z - 1)e^{e^z} = e^{-2z}\,e^{e^z}$

Hence, the solution of given equation is $y = y_c + y_p$

$y = c_1e^{-z} + c_2e^{-2z} - e^{-2z}e^{e^z}$, where $z = \log(x)$.

Exercise:

1. Solve $4x\frac{d^2y}{dt^2} + 4\frac{dy}{dt} - \frac{y}{x} = 4x.$

   Answer: $y = a\sqrt{x} + b\frac{1}{\sqrt{x}} + \frac{4}{15}x^2$

2. Solve $x^3\frac{d^3y}{dx^3} + 2x\frac{dy}{dx} - 2y = x^2\log x + 3x$

   Answer: $y = ax + x(b\,coslog\,x + c\,sinlog\,x) - \frac{1}{2}x^2 + 3x\log x$

3. Solve $(x^2D^2 + xD + 9)y = 3x^2 + sin(3\log x)$,

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where $D \equiv \frac{d}{dx}$.

Answer: $y = a\cos(3\log x) + b\sin(3\log x) + \frac{3}{13}x^2 - \frac{1}{6}\log x\cos(3\log x)$.

4. Solve $x^3\frac{d^3y}{dx^3} + 2x^2\frac{d^2y}{dx^2} + 2y = 10\left(x + \frac{1}{x}\right)$.

Answer:

$$y = \frac{c_1}{x} + x[c_2\cos(\log x) + c_3\sin(\log x)] + 5x + \frac{2}{x}logx.$$

5. Solve $x^3\frac{d^3y}{dx^3} + 3x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = x + logx$.

Answer:

$$y = \frac{c_1}{x} + \sqrt{x}\left[c_2\cos\frac{\sqrt{3}}{2}(\log x) + c_3\sin\frac{\sqrt{3}}{2}(\log x)\right] + \frac{x}{2} + logx.$$

Applications of differential equations:

A flexible spring is suspended vertically from a rigid support and then a mass is attached to its free end. Assuming that the mass vibrates, free of other external forces the displacement describes SHM. Fig (1)

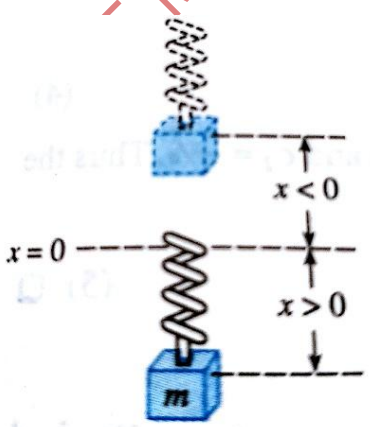

Fig (1)

1. The displacement 'x' of a particle executing simple harmonic motion (SHM) is given by $\ddot{x} = -n^2x$, where n^2 is a force constant. Solve this equation under the initial conditions $x = a$ and $\frac{dx}{dt} = 0$, at $t = 0$.

Solution: The given ODE is $({D_1}^2 + n^2)x = 0$,

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where $D_1 = \frac{d}{dt}$

The auxiliary equation and its roots are

$m^2 - n^2 = 0, m = \pm ni$

The general solution is given by

$x = c_1 \cos(nt) + c_2 \sin(nt)$, using the given conditions $x = a$ and $\frac{dx}{dt} = 0$, at $t = 0$ we find that $c_1 = a$ and $c_2 = 0$.

The displacement at any instant of time is given by $x = \text{a} \cos(nt)$

2. A vehicle supported by suspension systems involving coilovers(a vibration isolator with a damper inside a circular coil spring). The vehicle is rolling over a rough road with wavy road surface that could be described by a sine function, which results in a cyclic force of F(t) = $1200\, Sin(10t)$ acting on the vehicle. The differential equation for the amplitude of vibration of the vehicle mass $y(t)$ is given by $120\, y''(t) + 1920y'(t) + 12000y(t) = 1200 \sin(10t)$. What will be the position of the vehicle mass 2 seconds after the initiation of the vibration using initial conditions $y(0) = 0$ and $y'(0) = 5m/s$.

Solution:

The given differential equation is

$$120\, y''(t) + 1920y'(t) + 12000y(t) = 1200 \sin(10t)$$

Or $y''(t) + 16y'(t) + 100y(t) = 10 \sin(10t)$

With the initial conditions

$y(0) = 0$ and $y'(0) = 5m/s$

The auxiliary equation and its roots are

$m^2 + 16m + 100 = 0, m = -8 \pm 6i$

Hence the complementary function

$$y_c(t) = e^{-8t}(c_1 \cos 6t + c_2 \sin 6t)$$

The particular solution $y_p(t) = \frac{1}{d^2+16d+100}(10 \sin 10t)$

$$= 10\frac{1}{16d}(\sin 10t) = -\frac{1}{16}\cos 10t$$

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The complete solution is

$$y(t) = e^{-8t}(c_1 \cos 6t + c_2 \sin 6t) - \frac{1}{16}\cos 10t$$

Using the initial conditions,

We get $c_1 = \frac{1}{16}$ and $c_2 = \frac{11}{12}$.

Thus the particular solution is

$$y(t) = e^{-8t}\left(\frac{1}{16}\cos 6t + \frac{11}{12}\sin 6t\right) - \frac{1}{16}\cos 10t$$

3. The charge $q = q(t)$ on the capacitor in a LRC series circuit (Fig (2)) is governed by the equation $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t)$. Find the charge on the capacitor $q = q(t)$,

when L=0.25h, R=10Ω, C=0.001f, E(t) = 0, $q(0) = q_0$, $\frac{dq}{dt} = 0, for\ t = 0$.

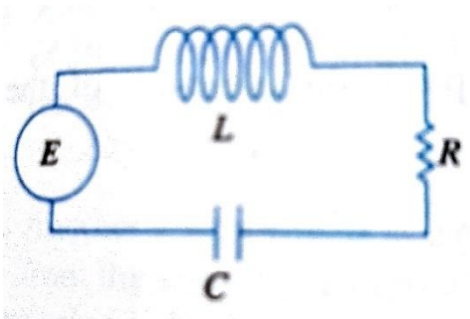

Fig (2)

Solution: The given ODE is $\frac{1}{4}\frac{d^2q}{dt^2} + 10\frac{dq}{dt} + 1000q = 0$

or $\frac{d^2q}{dt^2} + 40\frac{dq}{dt} + 4000q = 0$

The auxiliary equation and its roots are

$m^2 + 40m + 4000 = 0, m = -20 \pm 60i$

The general solution is given by $q = q(t)$

$q = e^{-20t}[c_1 \cos(60t) + c_2 \sin(60t)]$

Using the given conditions we get $c_1 = q_0$ and $c_2 = q_0/3$

The charge on the capacitor is given by

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$q = q(t) = e^{-20t}[q_0 \cos(60t) + \frac{q_0}{3} \sin(60t)]$.

4. Find the charge on the capacitor $q = q(t)$, in a LRC series circuit from $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t)$, when

L = (5/3)h, R = 10Ω, C = (1/30)f, E(t) = 300V, $q(0) = 0$, $\frac{dq}{dt} = 0$

Solution: The given ODE is

$$\frac{5}{3}\frac{d^2q}{dt^2} + 10\frac{dq}{dt} + 30q = 300$$

or

$$5\frac{d^2q}{dt^2} + 30\frac{dq}{dt} + 90q = 900.$$

The auxiliary equation and its roots are

$5m^2 + 30m + 90 = 0, m = -3 \pm 3i$

$$q_p = \left[\frac{1}{5D_1^{\,2}+30D_1+90}\right] 900e^{0t} = \frac{900}{90} = 10$$

The general solution is given by $q = q(t) = e^{-3t}[c_1 \cos(3t) + c_2 \sin(3t)] + 10$

Using the given conditions we get

$c_1 = -10$ and $c_2 = -10$

The charge on the capacitor is given by

$q = q(t) = 10 - e^{-20t}[\cos(3t) + \sin(3t)]$.

6. The radical displacement 'u' in a rotating disc at a distance r from the axis is given by $\frac{d^2u}{dr^2} + \frac{1}{r}\frac{du}{dr} - \frac{u}{r^2} + kr = 0$. Find the displacement if $u = 0$ at $r = 0$ and at $r = a$.

Solution: Here, we have $r^2\frac{d^2u}{dr^2} + r\frac{du}{dr} - u = -kr^3$

This is a homogeneous (Euler-Cauchy) equation

Put $r = e^z$, then $rDu = D_1u$, $r^2D^2u = D_1(D_1 - 1)u$, where $D = \frac{d}{dr}$.

The given equation reduces to $(D_1^{\,2} - 1)u = -ke^{3z}$

The auxiliary equation and its roots are

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$m^2 - 1 = 0, m = 1, -1. \therefore y_c = c_1 e^z + c_2 e^{-z}.$

Now $y_p = \frac{1}{D_1{}^2 - 1} - ke^{3z} = -k \frac{1}{3^2 - 1} e^{3z} = -\frac{k}{8} e^{3z}$

Hence, the solution of given equation is $y = y_c + y_p$

$y = c_1 e^z + c_2 e^{-z} - \frac{k}{8} e^{3z}$, where $z = \log(r)$.

Exercise:

1. A body executes damped forced vibrations given by the equation $\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + b^2 x = e^{-kt} sin\omega t$. Solve the equation for both the cases, when $\omega^2 \neq b^2 - k^2$ and when $\omega^2 = b^2 - k^2$.

Solution: If $\omega^2 \neq b^2 - k^2$

$x = e^{-kt}\left[c_1 cos\sqrt{b^2 - k^2}t + c_2 sin\sqrt{b^2 - k^2}t\right] + \frac{e^{-kt}}{b^2 - k^2 - \omega^2} sin\omega t.$

If $\omega^2 = b^2 - k^2$

$x = e^{-kt}[c_1 cos\omega t + c_2 sin\omega t] - \frac{t}{2\omega} e^{-kt} cos\omega t.$

Video links:

1. https://www.youtube.com/watch?v=V9bl02Ffo_o
2. https://youtu.be/uI2xt8nTOlQ?si=RJzVW7J5MeRMOAba
3. https://youtu.be/B1ZB8nvWgcw?si=ocpwu_6QFxmWlQyC

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